

TUZA’S CONJECTURE FOR SPLIT GRAPHS WITH AN EIGHT-VERTEX CLIQUE PART AND TWO NEIGHBORHOOD TYPES

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ABSTRACT. For a graph G , let $\nu_{\Delta}(G)$ be the largest number of pairwise edge-disjoint triangles and let $\tau_{\Delta}(G)$ be the smallest number of edges meeting every triangle. Tuza conjectured that $\tau_{\Delta}(G) \leq 2\nu_{\Delta}(G)$ for every graph. We prove the conjecture for every split graph with a specified split partition $V(G) = C \dot{\cup} I$, $|C| = 8$, for which the triangle-active vertices of I have at most two distinct neighborhoods in C . The result allows arbitrary multiplicities of the two types. The infinite statement is reduced to 44,702 canonical cases by an exact multiplicity-truncation lemma and symmetry. Each case carries an explicit triangle cover and an explicit edge-disjoint triangle packing, checked by an independent standard-library verifier. We also give exact cover and packing reductions for arbitrary split graphs and a maximum-cut criterion that settles, at $|C| = 8$, every instance whose independent-side neighborhood 2-shadow has at most seven edges. The unrestricted split-graph and interval-graph cases remain open.

1. INTRODUCTION

For a finite simple graph G , a *triangle packing* is a family of pairwise edge-disjoint triangles, and a *triangle cover* is a set of edges meeting every triangle. Write $\nu_{\Delta}(G)$ and $\tau_{\Delta}(G)$ for their maximum and minimum cardinalities. Tuza’s conjecture asserts

$$(1) \quad \tau_{\Delta}(G) \leq 2\nu_{\Delta}(G)$$

for every G [5]. It was posed at the 1981 Eger colloquium and has remained open in general.

Split and interval graphs remain important unresolved subclasses [3]. Some adjacent frontiers are known. The conjecture holds for threshold graphs [1], for K_8 -free chordal graphs [2], for an even-sided co-chain subclass of interval graphs [3], and for split graphs of order n and minimum degree at least $3n/5$ [4]. Thus an eight-vertex clique part is the first partition size not covered merely by the K_8 -free chordal result. Notice that

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a specified split partition with $|C| = 8$ can still have clique number larger than eight when an independent-side vertex is complete to C .

Our principal result is the following infinite, computer-assisted theorem. Vertices of I with at most one neighbor in C lie in no triangle and are called *inactive*.

Theorem 1.1. *Let G be a split graph with a specified split partition $V(G) = C \dot{\cup} I$ and $|C| = 8$. After deleting the inactive vertices of I , suppose that the sets $N_G(x) \cap C$, $x \in I$, take at most two distinct values. Then G satisfies (1).*

The proof has two logically separate parts. First, an exact lemma truncates the multiplicity of every neighborhood type at $\binom{|C|}{2}$. Second, symmetry reduces the one- and two-type instances at $|C| = 8$ to 44,702 records. Every record stores a feasible cover F and a feasible packing $\mathcal{P}$ with $|F| \leq 2|\mathcal{P}|$. The independent verifier reconstructs all graph edges and all relevant triangles; it does not call the discovery program or an optimization solver.

We also prove exact cover and packing formulas for arbitrary split graphs and a maximum-cut sufficient condition. At $|C| = 8$ the latter gives a purely human-verifiable theorem for arbitrary neighborhood types whenever the independent-side neighborhood 2-shadow has at most seven edges.

Theorem 1.1 is a partial result: it proves neither all split graphs nor all interval graphs. The literature search supporting the novelty and scope statements was last updated on August 17, 2026.

2. EXACT REDUCTIONS FOR SPLIT GRAPHS

Fix a split partition $V(G) = C \dot{\cup} I$. Put $N_x = N_G(x) \cap C$ for $x \in I$, and identify $G[C]$ with K_C .

Lemma 2.1 (Exact cover reduction). *For $X \subseteq E(K_C)$, let*

$$H_x(X) = (N_x, E(K_{N_x}) \setminus X).$$

Then

$$(2) \quad \tau_\Delta(G) = \min_{\substack{X \subseteq E(K_C) \\ K_C - X \text{ triangle-free}}} \left(|X| + \sum_{x \in I} \tau_v(H_x(X)) \right),$$

where τ_v is the ordinary vertex-cover number.

Proof. Let F be a triangle cover and $X = F \cap E(K_C)$. Since F meets every triangle inside C , the graph $K_C - X$ is triangle-free. For $x \in I$, set $Z_x = \{u \in N_x : xu \in F\}$. If $uv \in E(H_x(X))$, then $uv \notin X$, so the triangle xuv forces $u \in Z_x$ or $v \in Z_x$. Hence Z_x is a vertex cover of $H_x(X)$, giving the lower bound in (2).

Conversely, take feasible X and a minimum vertex cover Z_x of every $H_x(X)$. Then

$$X \cup \{xu : x \in I, u \in Z_x\}$$

meets all clique triangles because $K_C - X$ is triangle-free. It also meets every triangle xuv : either $uv \in X$, or $uv \in E(H_x(X))$ and one of u, v belongs to Z_x . This proves the reverse bound. $\square$

Equivalently, writing $H = K_C - X$ and $q = \binom{|C|}{2}$, Lemma 2.1 gives

$$(3) \quad \tau_\Delta(G) = \min_{\substack{H \subseteq K_C \\ H \text{ triangle-free}}} \left(q - |E(H)| + \sum_{x \in I} (|N_x| - \alpha(H[N_x])) \right).$$

Lemma 2.2 (Exact packing reduction). *For every $x \in I$, choose a matching $M_x \subseteq E(K_{N_x})$, with the edge sets M_x pairwise disjoint, and put $P = \bigcup_x M_x$. Then*

$$(4) \quad \nu_\Delta(G) = \max_{(M_x)} \left(\sum_{x \in I} |M_x| + \nu_\Delta(K_C - P) \right).$$

Proof. An edge $uv \in M_x$ represents the triangle xuv . The matching condition prevents two triangles with center x from sharing a spoke; pairwise edge-disjointness prevents reuse of a clique edge. A clique-triangle packing in $K_C - P$ may then be added. Conversely, the clique edges of packed triangles centered at a fixed x form such a matching, and the remaining packed triangles lie in $K_C - P$. $\square$

For $S \subseteq C$, let $m_S = |\{x \in I : N_x = S\}|$.

Lemma 2.3 (Multiplicity truncation). *Let $q = \binom{|C|}{2}$. Replacing every m_S by $\bar{m}_S = \min\{m_S, q\}$ leaves both $\tau_\Delta(G)$ and $\nu_\Delta(G)$ unchanged.*

Proof. Every triangle meeting I consumes a clique edge. Thus at most q copies of any one type can occur in a packing, and any used copies can be relabeled. This proves the packing assertion.

For covering, combine equal types in (3) and put

$$d_S(H) = |S| - \alpha(H[S]), \quad \Phi_m(H) = q - |E(H)| + \sum_S m_S d_S(H).$$

If the optimum for the truncated multiplicities equals q , the empty graph H has value q for the original multiplicities as well, so both optima are q . Otherwise let H be a truncated minimizer of value less than q . For every saturated type $\bar{m}_S = q$ we must have $d_S(H) = 0$, since otherwise that term alone is at least q . Hence the same H has exactly the same value for the original multiplicities. Monotonicity in every m_S supplies the reverse inequality. $\square$

3. A MAXIMUM-CUT CRITERION

Define the *independent-side neighborhood 2-shadow* Q on C by

$$(5) \quad uv \in E(Q) \iff \text{there exists } x \in I \text{ with } u, v \in N_x.$$

The complement below is taken inside K_C .

Theorem 3.1. *Let $k = |C|$ and $q = \binom{k}{2}$. Then*

$$(6) \quad \tau_{\Delta}(G) \leq q - \text{MaxCut}(\overline{Q}).$$

Consequently, if

$$\text{MaxCut}(\overline{Q}) \geq q - 2\nu_{\Delta}(K_k),$$

then G satisfies Tuza's conjecture.

Proof. For a cut $C = A \dot{\cup} B$, select every clique edge internal to A or B , together with every crossing edge belonging to Q . Every clique triangle has two vertices on the same side. A triangle xuv is also met: if u, v are on opposite sides, then $uv \in E(Q)$ by definition. The number of selected edges is $q - e_{\overline{Q}}(A, B)$. Maximizing the omitted crossing edges proves (6). The final assertion follows from $K_C \subseteq G$, and hence $\nu_{\Delta}(K_k) \leq \nu_{\Delta}(G)$. $\square$

Corollary 3.2. *If $|C| = 8$ and the shadow Q has at most seven edges, then G satisfies Tuza's conjecture, with no restriction on the number or form of its neighborhood types.*

Proof. Choose a uniformly random balanced 4–4 cut of C . Each edge of Q crosses with probability $16/28$, so some balanced cut has at most four crossing edges of Q . The cover used in Theorem 3.1 then has at most $2\binom{4}{2} + 4 = 16$ edges. On vertices $0, \dots, 7$, the triangles

$$017, \quad 023, \quad 045, \quad 124, \quad 136, \quad 256, \quad 357, \quad 467$$

are pairwise edge-disjoint. Thus $\nu_{\Delta}(G) \geq 8$, and the result follows. $\square$

4. THE EXACT CERTIFICATE

We now prove Theorem 1.1. One active type is classified by its size $2, \dots, 8$. Two distinct active types $S, T \subseteq C$ are classified, up to a permutation of C and an exchange of S, T , by

$$(7) \quad (u, a, b, o) = (|S \cap T|, |S \setminus T|, |T \setminus S|, |C \setminus (S \cup T)|),$$

with $a \leq b$. There are exactly 64 two-type classes. When $a = b$, type exchange also permits $m_S \leq m_T$.

By Lemma 2.3, it is enough to check multiplicities from 1 to 28. The canonical scope therefore contains $7 \cdot 28 = 196$ one-type records and

$$49 \cdot 28^2 + 15 \cdot \binom{29}{2} = 44,506$$

two-type records, for 44,702 records in total.

For each record, the certificate contains an explicit triangle cover and an explicit edge-disjoint triangle packing. A cover is serialized as a set of clique edges together with one spoke-endpoint mask for each type. A packing is serialized triangle by triangle, distinguishing clique triangles from triangles centered at a named independent vertex. The discovery program constructs packings from ordered round-robin matching factors of K_S and K_T , tries

both type orders, and fills unused clique edges by an exact K_8 triangle-packing search. This construction is outside the trust boundary: only the stored witnesses matter.

For clarity, type compression does not identify different graph edges. If a type S has multiplicity m , the stored endpoint mask $Z_S \subseteq S$ is applied separately to all m copies, contributing $m|Z_S|$ distinct spokes. The same mask meets every remaining centered triangle because all copies have the same clique neighborhood.

Proposition 4.1. *Every one of the 44,702 canonical records contains a triangle cover F and an edge-disjoint triangle packing $\mathcal{P}$ satisfying $|F| \leq 2|\mathcal{P}|$. The minimum certified slack $2|\mathcal{P}| - |F|$ is 4.*

Exact verification. The independent verifier uses only the Python standard library and imports nothing from the generator or the exact solver. It independently re-enumerates the 64 Venn classes and all 44,702 records; rejects missing, reordered, malformed, or trailing data; expands each packed triangle into its three graph-edge tokens and checks pairwise edge-disjointness; and checks every clique triangle and independent-centered triangle against the stored cover. Finally it checks $|F| \leq 2|\mathcal{P}|$.

The canonical uncompressed JSONL data have SHA-256

09966099c6a19214ff585624cb0d1de8ea8c54406783701f4901bb0641a60bc8.

The compressed certificate, generator, independent verifier, and manifest have respective SHA-256 values

b4f1343c4aaa0b8f672b741eb69e61324d52997558abed3697abb09eb8ccdde5,
b186a126320bc0fcf1f97bfcf97e5e80208cc3423781c6c5f4a8982f9a042d03,
1dadda8af5be397fa768fca1a020acf97c6bce2f81cfd7b3979733d4eda0723f,
45d4b143b9b9eb87bc2e8b75916f4ce857a93cef48066a51c7697ad70eca1325.

Two consecutive verification reports are byte-identical, with SHA-256

b1e28e146088a76d5c5389b3522a04a09fbb011e2b67b7a595cacb1b08140514.

□

Proof of Theorem 1.1. Delete inactive vertices, which occur in no triangle. If there are no active types, the triangle core is K_8 . The eight triangles displayed in the proof of Corollary 3.2 form a packing, while the 12 edges internal to a balanced 4-4 partition cover all clique triangles; hence $\tau_\Delta(K_8) \leq 12 \leq 16 \leq 2\nu_\Delta(K_8)$.

Suppose there are one or two active types. Lemma 2.3 replaces each multiplicity by its minimum with 28 without changing either parameter. Relabeling C and, when applicable, exchanging the two types puts the instance into exactly one canonical record. Proposition 4.1 gives

$$\tau_\Delta(G) \leq |F| \leq 2|\mathcal{P}| \leq 2\nu_\Delta(G).$$

□

Remark 4.2. At multiplicity 28, the verifier also checks that every stored cover uses no spoke of the saturated type. Thus the same witnesses extend directly after adding twins. This check is redundant because Lemma 2.3 already proves the extension analytically.

5. REPRODUCIBILITY AND LIMITATIONS

From the project root, run

```
python3 experiments/ilp_breaker/verify_venn_k8.py
```

for full witness verification, and

```
PYTHONPATH=src:experiments/ilp_breaker python3 -m pytest -q
```

for the tests. On the pinned artifact both independent verification runs accepted all 44,702 records, and all eight project tests passed. The machine-readable theorem scope is stored separately from the certificate. The verifier checks feasibility, disjointness, completeness of the enumerated scope, the inequality, and digests; it does not assume that the stored cover or packing is optimal.

No floating-point computation enters the proof. The certificate uses exact integer and bit-mask arithmetic. This is not a Lean or other proof-assistant formalization: the trust base consists of the finite reduction, the Python interpreter and standard library, the independent verifier, and the stored witnesses.

The unrestricted split-graph and interval-graph cases remain open, as does the case of three or more unrestricted neighborhood types. The reductions in Section 2 and the shadow criterion in Section 3 provide routes toward those extensions.

COMPUTATIONAL AND AI DISCLOSURE

OpenAI’s Codex system assisted with route exploration, program construction, generation of exact certificates, adversarial checking, literature searches, and language editing. The mathematical statements, executable artifacts, and final manuscript remain the responsibility of the author. No claim of formal peer review or proof-assistant verification is made.

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